

6.3 - Detailed CSC / CSU Descriptions

The application is composed of the following CSCs:

5.2.1 - Data Contribution CSC

Application component dedicated to reporting microforest data from the field.

5.2.1.1 - Log In/Sign Up CSU

- 5.2.1.1.1 - Application Title (Display title of application)

- 5.2.1.1.2 - Log In button

- 5.2.1.1.3 - Sign Up button

- 5.2.1.1.4 - Log In/Sign Up fields (username, password)

5.2.1.2 - Plant Data Form Tab CSU

- 5.2.1.2.1 - Data fields (Plant ID, plant health, plant height)

- 5.2.1.2.2 - QR Code Scanner interface

- 5.2.1.2.3 - Submit button

5.2.1.3 - Registered Forests Tab CSU

- 5.2.1.3.1 - Registered forests list

- 5.2.1.3.2 - Register with new forest Button

5.2.2 - Administrator (Ranger) Portal CSC

Application component dedicated to enabling admins to manage their microforest data securely.

5.2.2.1 - Admin Sign In CSU

- 5.2.2.1.1 - Log In/Sign Up fields (email, username, password)

- 5.2.2.1.2 - Forgot password button

5.2.2.2 - Admin Dashboard CSU

- 5.2.2.2.1 - Analytics Window (headline statistics and data summary)

- 5.2.2.2.2 - Users Window (lists users that have contributed)

- 5.2.2.2.3 - Data Window (lists granular data that has been gathered)

- 5.2.2.2.4 - Forest selector (dictates which forests' data is shown)

- 5.2.2.2.5 - Download as CSV button (allows local download of logs)

- 5.2.2.2.6 - Sign-out Button

5.2.3 - Microforest Connect CSC

Visual web interface displaying microforest locations and environmental impact.

- 5.2.3.1 - Map CSU (interactive map of microforests)

- 5.2.3.1.1 - Microforest Pings (locations of microforests on map)

- 5.2.3.2 - Microforest Details CSU (details sheet about microforests)
- 5.2.3.2.1 - Microforest Title

- 5.2.3.2.2 - Microforest stewards (responsible organization(s))

- 5.2.3.2.3 - Microforest scale (size)

- 5.2.4 - Backend Database CSC

The data storage and manipulation system that serves as the project's central hub.

- 5.2.4.1 - Forests Database CSU

- 5.2.4.1.1 - Forest Metadata (location, administrators, etc.)

- 5.2.4.1.2 - Plant Data (individual logs of plant health statuses)

- 5.2.4.2 - Users Database CSU

- 5.2.4.2.1 - User Metadata (name, email, username)

- 5.2.4.2.2 - Registered Forests (list of forests the user contributes to)

6.3.1 - Detailed Class Description

The following sections provide the details of all core classes utilized in the Ascot Project's top-down architectural design. The majority of CSUs above are User Interface component classes, so this section will be focusing on the classes that form the core functionality of the project regarding plant report data, the user class, and administrative permissions. It's important to note that both the Steward portal and the Data Contribution app make use of the same user objects(class 6.3.1.1).

- 6.3.1.1 - users

- The users class represents a registered individual using the application, storing personal metadata and maintaining their access links to various forests.*

- 6.3.1.1.1 - Fields

- 6.3.1.1.1.1 - userID: UUID (Unique identifier assigned to the user)

6.3.1.1.1.2 - fullname: string (The user's full name)

6.3.1.1.1.3 - email: string (The user's contact email)

6.3.1.1.1.4 - authorizedForests: list[] (A list associating the user to the specific microforests they are authorized to contribute to)

6.3.1.2 - forest

The forest class represents a specific microforest project, containing overarching information about the site as well as aggregating all logged plant data.

6.3.1.2.1 - Fields

6.3.1.2.1.1 - forestName: string (The designated title of the microforest)

6.3.1.2.1.2 - metadata: JSON (Site details, including geographic coordinates) 6.3.1.2.1.3 - administrators: list[] (A list of admins authorized to manage this forest)

6.3.1.2.1.4 - plantData: plantDataPoint[] (An array containing the historical health logs recorded at this forest)

6.3.1.3 - plantDataPoint

The plantDataPoint class represents a single log or field report submitted for a specific plant.

6.3.1.3.1 - Fields

6.3.1.3.1.1 - id: UUID (Unique identifier for the report instance)

6.3.1.3.1.2 - author: UUID (Reference to the user who submitted the report)

6.3.1.3.1.3 - date: date (The date the record was submitted)

6.3.1.3.1.4 - dead: boolean (Status indicator of whether the plant is dead)

6.3.1.3.1.5 - evidenceOf: evidenceStates (Nested class containing specific observational conditions)

6.3.1.3.1.6 - plantHealth: enumeration (Overall health condition restricted to GOOD, FAIR, or POOR)

6.3.1.3.1.7 - plantHeight: float (The recorded height of the plant)

6.3.1.3.1.8 - plantID: string (The unique ID for the plant, typically read via QR code)

6.3.1.3.1.9 - speciesID: string (Identifier for the plant's species)

6.3.1.4 - evidenceStates

A substructure utilized within plantDataPoint to capture granular boolean conditions observed during data collection.

6.3.1.4.1 - Fields

6.3.1.4.1.1 - dryingOut: boolean

6.3.1.4.1.2 - predation: boolean

6.3.1.4.1.3 - herbivory: boolean

6.3.1.4.1.4 - flowering: boolean

6.3.1.4.1.5 - seeding: boolean

6.3.2 - Detailed Interface Description

The major software interactions within the Ascot system rely centrally on the Backend Database to synchronize the mobile and web applications without them needing to interact directly.

6.3.2.1 - Data Contribution Mobile App and Backend Database

The mobile application interfaces directly with the Backend Database by submitting real-time logs. **When a contributor submits a form, the app performs a write operation to the Forests Database, generating a historical log indexed by the Plant ID.** Simultaneously, the database supplies the app with the necessary read data to populate the user's list of "Registered Forests" based on their authentication profile.

6.3.2.2 - Administrator Portal and Backend Database

The Administrator Portal relies on database reads to populate the Analytics, Users, and Data windows on the admin dashboard. **Any administrative modifications, such as managing users or deleting erroneous reports, are propagated back to the Backend Database** to guarantee system-wide integrity. This ensures that granular plant logs are readily accessible for CSV exports.

6.3.2.3 - Microforest Connect Web Interface and Backend Database

The Microforest Connect Web Interface **functions strictly as a read-only consumer** of the database. It queries and compiles forest metadata alongside calculated health statistics to render interactive map pings and public fact sheets. This ensures the public only sees validated information synchronized with the Admin Portal.

6.3.2.4 - Data Contribution App and Device Hardware

To operate successfully in outdoor environments, the Data Contribution App must access native mobile hardware. This interface includes **utilizing the mobile camera to operate the QR Code Scanner** for automated data entry, depending on a high-brightness screen to counteract high-glare environments, and leveraging 4G/5G connections for real-time database syncing.

6.3.3 - Detailed Data Structure Description

6.3.3.1 - Application Data Collections

6.3.3.1.1 - forests list - A list structure mapping the entire array of registered forest environments stored within the application's backend.

6.3.3.2 - Data Formats

6.3.3.2.1 - metadata JSON - A **JSON formatted object** utilized within the forest class. It manages and packages various unstructured details about a site, such as its geographic coordinates.

6.3.3.2.2 - plantHealth enumeration - A strict enumerated type representing the overall visual health of a specific plant. Valid values are restricted to GOOD, FAIR, or POOR.

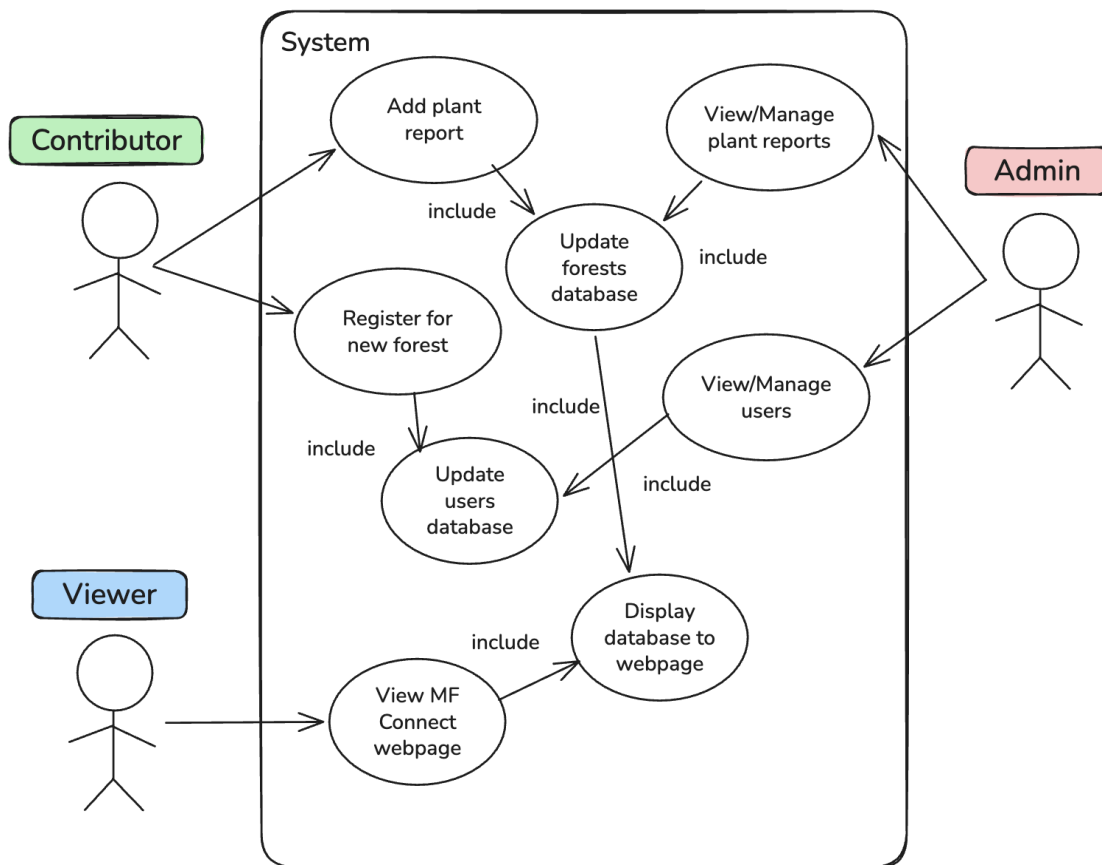
6.3.3.3 - Relational Structures

6.3.3.3.1 - authorizedForests list - A list maintained within the users database schema. It serves as **a relational link** connecting user profiles to the precise list of microforest databases they are allowed to report data to.

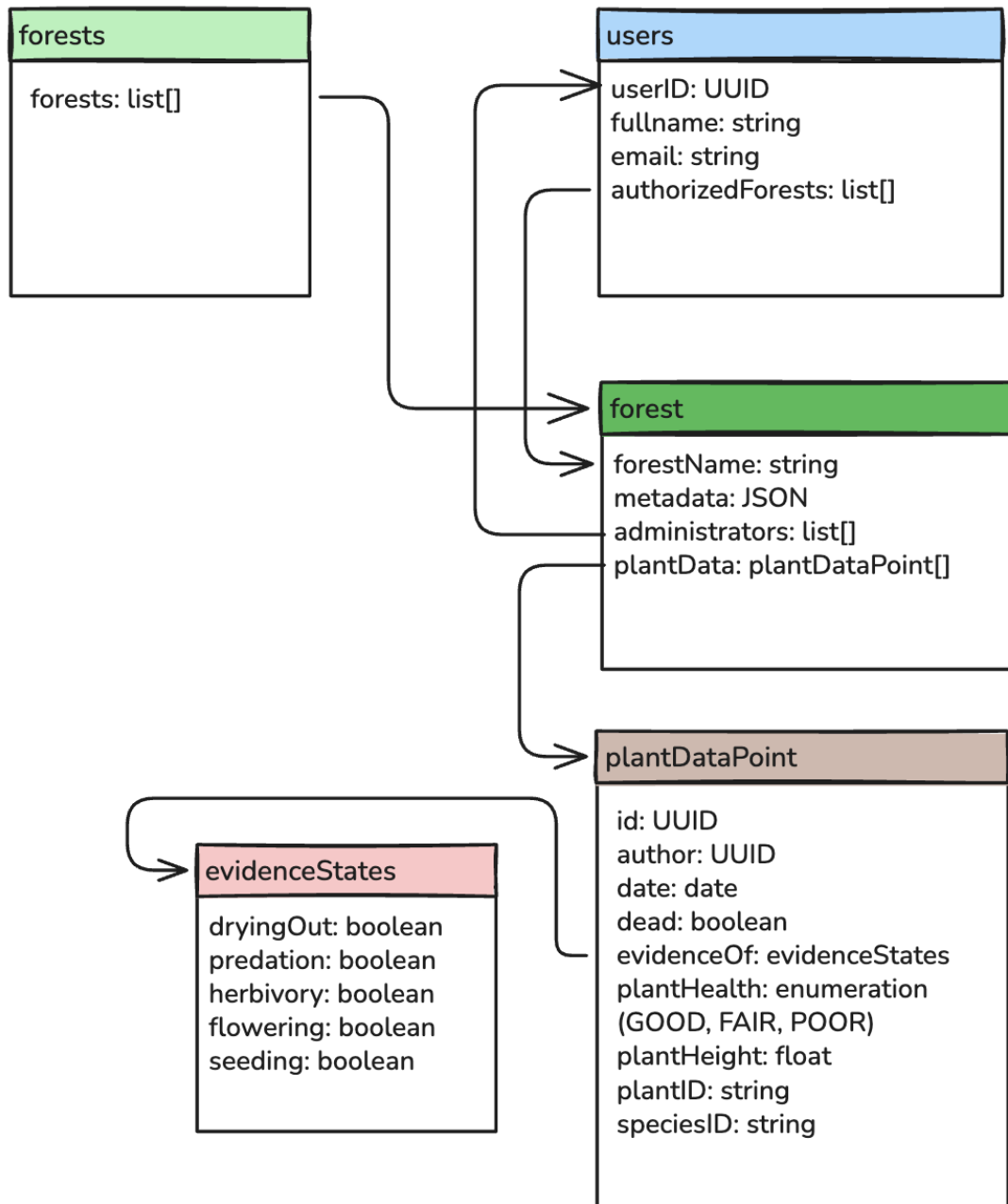
6.3.3.3.2 - plantData array - An array data structure built into the forest database schema. It stores all plantDataPoint entries, acting as a historical log mapping the respective microforest to all its individual plant metrics.

6.3.4 - Detailed Design Diagrams

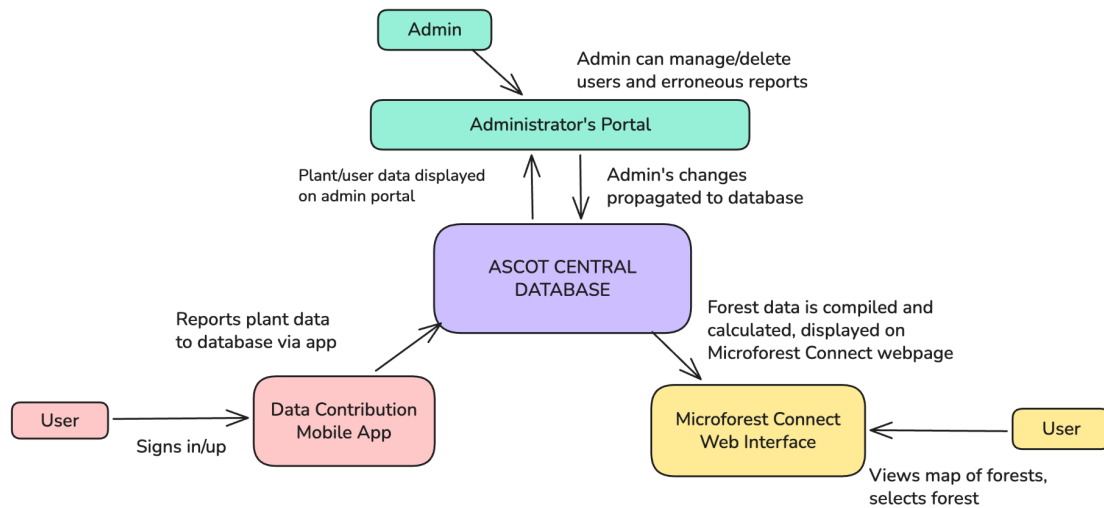
Use Case Diagram



Top-Down Class Diagram

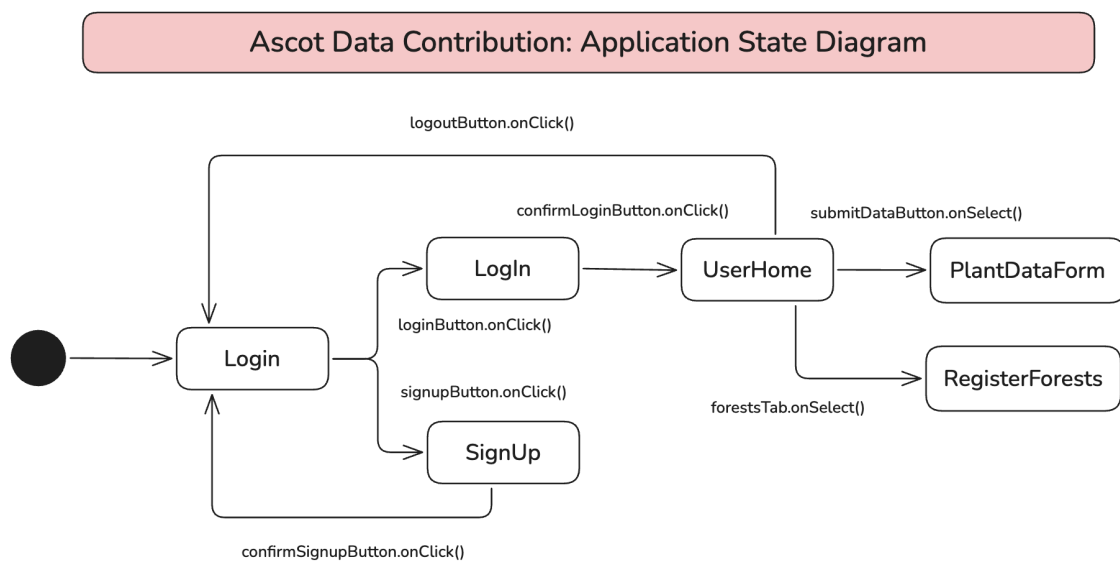


Component Diagram



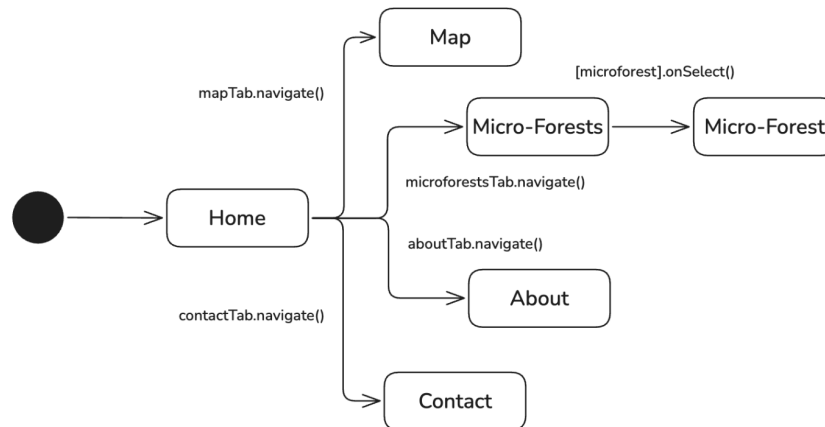
State Diagrams

Ascot Data Contribution



Ascot Microforest Connect

Ascot Micro-Forest Connect: Application State Diagram



Ascot Steward(Administrator) Portal

Ascot Data Contribution: Application State Diagram

